Homework 3: PySide6 GUI

For this assignment, you will be using **PySide6** to create a GUI application for an enhanced image search engine. The GUI will contain at least the following widgets: `QLineEdit`, `QComboBox`, and `QPushButton`.

The user will enter a search term into a `QLineEdit` widget. A `QComboBox` will allow the user to select one of the following image manipulations (all covered in lecture/lab):

- Sepia
- Negative
- Grayscale
- Thumbnail
- None

> Your use of `Pillow` for these image manipulations should be restricted to the following methods: `new()`, `open()`, `getdata()`, `putdata()`, `getpixel()`, `putpixel()`, and `save()`. For the thumbnail, create an image which is half the width and half the height of the full-size image. (Again, all of these techniques were covered in lecture/lab.)

Once the user presses the "Search" button (a `QPushButton` widget), your GUI should display the image result with the best match using PySide6. (See "Search Rules" below.) This can be done in the existing window or a new window, but should use PySide6 (**not** Pillow) to display the image.

The full-size images are provided here. Do not change the names of the images or the `image_info.py` file.

Image Metadata

The image metadata file, `image_info.py`, is provided here. It is in the form of a list of dictionaries. For reference, here is the first element of the list (which itself is a dictionary):

```
{
    "id" : "3260578960_43b778eb8b_c",
    "title" : "Matmata", "flickr_user" : "Sek Keung Lo",
    "tags" : ["Tunisia", "village", "cave", "dwelling", "lifestyle"],
    "url" : "https://www.flickr.com/photos/losk/3260578960"
}
```

The `id` corresponds with the file name. For example, `3260578960_43b778eb8b_c` corresponds to image `3260578960_43b778eb8b_c.jpg` (which is stored in the provided `hw3_images` folder).

Search Rules

Your program should be able to perform a single word search on the image titles and tags provided in `image_info.py`.
Example search terms to test: "gardens", "vessel", "memorial", "airport", "otter"

A few more details:

- Given a search term, search through the title **and** tags (provided in the `image_info` list) looking for matches. Your program should maintain a record of the number of matches for each image.
- The image with the highest number of matches should be returned.
- The search should not be case sensitive.
- Your code will not be tested on search terms where there's a "tie" (same number of matches) or where there are partial matches ("garden" in "gardens", etc.).

If a search comes up with no results, the default image should be `no_results.jpg`:

(This image is also provided in the `hw3_images.zip` file.)

functions.py and gui.py

Your search function, `my_search()` and image transformation functions should be stored in a file called `functions.py`. Your `my_search()` function should take a **single'** string argument and return an image id or "no results".

You will import `functions.py` into your GUI code file, `gui.py`. For example, for your search function, `my_search()`, your import will be:

```
from functions import my_search
```

Important

Do not change the `image_info.py` file. Place it in the same directory as `gui.py` and `functions.py` and import it with the following code:

```
from image_info import image_info
```

Do not change the names of any of the image files. The images should be in a directory called `hw3_images`.

To receive full credit, your `gui.py` file must include the following import:

```
from __feature__ import snake_case, true_property
```

Deliverable

Submit your source code files (`gui.py` and `functions.py`), as well as a screenshot of your GUI (including your Desktop). Each source code file should contain a header comment containing essential information (e.g., your name, the class, the date, brief description).

Grading

GUI & Image Manipulations — 45 points

- Correct widgets included
- Image manipulations properly implemented
- Images properly displayed

Search — 45 points

- Return the image with the most matches (in both title and tags). item
- Not case sensitive
- If no match is found, should show no_results.jpg (with no image manipulation performed)

Deliverable — 10 points

- Header comments (in `gui.py` and `functions.py`): 4/4
- Screenshot: 6/6

Lastly, do your own work! If your code bears excessive resemblance to another student's code, you will both receive zero credit and will be reported to the Department Chair.

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